

TRADITIONAL AND BIG WALL FREE CLIMBING



Traditional climbing can take you to magnificent locations. Mike Anderson on the upper reaches of Half Dome during the First Free Ascent of *Arcturus*, 5.13b, Yosemite, CA.  Andrew Burr

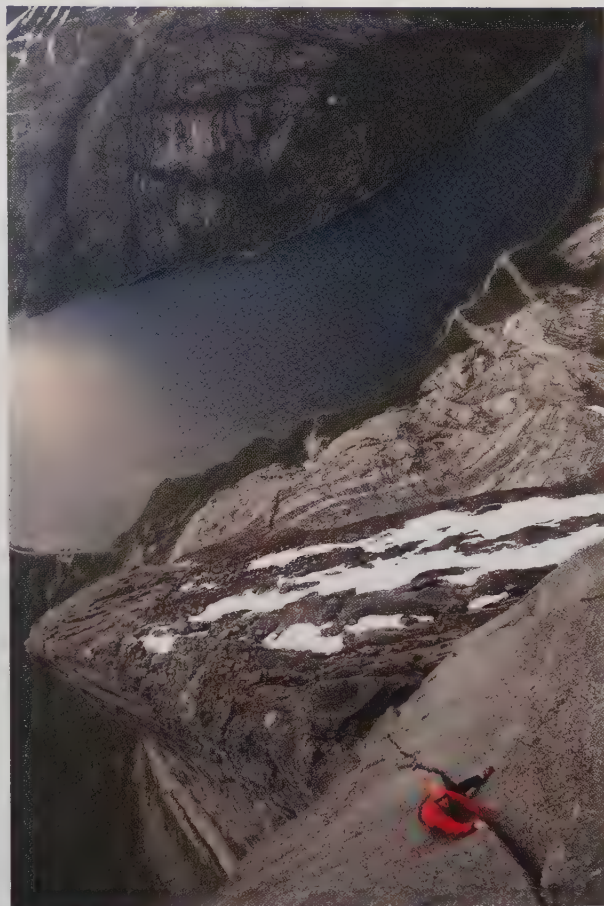
CHAPTER 14

“The magnificent beauty and historic significance of [The Nose], as well as my own efforts to free it, then later freeing it in a day, made this ascent the most meaningful achievement of my entire climbing career.” — Lynn Hill¹

Traditional climbing skills provide access to some of the most spectacular and inspiring features on the planet. From soaring shields of flawless Sierra granite to gravity-defying towers of Wingate sandstone, the ability to perform on gear-protected terrain opens the door to a magnificent world of limitless adventure. While the intricacies of gear placement are beyond the scope of this book, there are technical climbing skills and physical-fitness requirements unique to traditional (trad) climbing that can make a significant difference on the sharp end. Furthermore, besides its intrinsic value, trad climbing can also reveal gaps in a climber’s capabilities, and thus provide an excellent means to improve general climbing weaknesses.

THE UNIQUE REQUIREMENTS OF TRADITIONAL CLIMBING

Due to the differences that often exist between sport and trad climbs, a slightly modified training approach should be used. While traditional routes can take nearly any form, this discussion focuses primarily on the most archetypical trad routes: granite and sandstone gear-protected climbs that are not very steep. These types of trad climbs differ from the typical sport climb in many important ways. Training adjustments can be made to better prepare for such a goal trad route. These general differences include:



Trad routes can be much longer than the typical sport pitch, requiring modified training. Rob Pizem high over the fjord on Stetind, Norway.

Andrew Burr

- **Length:** A sport pitch rarely exceeds half a rope length (30m) and is usually much shorter, while some trad pitches go for an entire rope length (60m) and can be much more physically repetitive.
- **Duration:** Regardless of the pitch length, trad routes often require more time to redpoint due to the need to place protection and the

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intricate hand placements required for jamming cracks.

- **Angle:** Trad climbs are generally vertical to slabby (step back from the cliff and it's clear that most of those Indian Creek splitters are less than vertical).
- **Pace:** The need to place protection and fiddle with jams slows the trad-climbing pace.
- **Hold type and size:** The jams afforded on a trad climb require specialized techniques to exploit, and may be painful, or even injurious.
- **Duty cycle:** Handholds will generally need to be gripped much longer than those on a typical sport climb, but they're usually better holds. The rest period between holds is also longer owing to the need to carefully place intricate jams.
- **Rests:** The occasional "bomber" jam means that rests will likely be more frequent, but they may be more complex and less amenable to alternating-hand shakeouts.
- **Gear:** Removable protection adds weight to the climber and slows him down. It may also increase leading anxiety and (justifiably in some cases) reduce the willingness to go for it.

- **Onsight:** Because of historical practices, some climbers emphasize a ground-up or onsight-preferred approach to trad climbs. Big walls or multipitch climbs often necessitate such an approach.
- **Shoes:** Certain trad climbs may call for specialized climbing shoes (such as a very stiff-soled shoe) that otherwise hamper general footwork.
- **Conditions:** Trad climbs are often located in extreme environments with imperfect conditions. Furthermore, on multipitch climbs, it may not be possible to time the climbing so that the hard pitches are climbed under the best conditions.

Trad climbs generally share many traits with vertical face climbing, which makes them good substitutes for each other for training and/or practice. Trad and face climbing both benefit from strong, precise footwork by allowing the feet to support a large portion of the climber's weight. They both tend to have small face holds that require precision and power to exploit. Additionally, with the obvious exception of the perfectly continuous enduro cracks of the Colorado Plateau,



Trad routes typically hinge on the climber's ability to execute short, technical boulder problems, where power and footwork are paramount. Jean-Pierre Ouellet finding a *Home on the Range*, 5.13, Vedauwoo, WY.

📷 Andrew Burr

most trad climbs consist of one or more bouldery cruxes separated by generally good rests. While endurance is a factor on these climbs, they rarely require a *power-endurance* sprint typical of many overhanging sport climbs. Instead, the redpoint most often hinges on the ability to execute boulder-problem cruxes above gear, and recover at the intermittent rests afforded by a good jam or large foothold. These critical characteristics dictate how best to train for these types of trad climbs.

TRAINING ADJUSTMENTS FOR

TRAD CLIMBING

To maximize trad-climbing performance, the Rock Prodigy program should be adjusted to account for the unique requirements of trad climbs. The training should be tweaked to improve the climber's ability to execute technical, vertical boulder problems and to sustain long-duration redpoint or onsight efforts. Therefore, notwithstanding a specific goal route with more definitive characteristics, a climber training for a generic trad-climbing goal should increasingly empha-

Hard Trad...Or Is It Sport?

"I haven't learned nearly as much from cracks that I can apply to face as I have from face that I can apply to cracks. Sport climbing gives you a new repertoire of moves and ways to position your body. For me, faces used to be the crux — now they're an opportunity." — Steve Petro⁴

As routes increase in difficulty, the apparent distinctions between trad and sport climbs begin to fade. With few exceptions, as cracks get more difficult, they become narrower, incipient, and less secure. The crack becomes less useful for upward progress so the fundamental jamming skills that are essential on a moderate crack climb become less critical to success. Instead, difficult cracks rely more on face holds and face-climbing techniques, and the crack itself, rather than being the star of the show, is just another member of the supporting cast, contributing its poor features for face climbing. It is also common for so-called "trad" routes to sprout more fixed protection (which may include fixed pins, nuts, cams, bolts, etc) as the difficulty increases, further blurring the line between "sport" and "trad."

These distinctions can be left to the media for now, but as goal-oriented athletes, our focus should remain on what, if any, changes should be made to our training to prepare for a difficult trad goal route. Fortunately, this merger of climbing styles toward the top of the grade scales actually simplifies the job of selecting and accomplishing the proper training. As you rise through the grades, it becomes more likely that your typical training approach to sport climbing will be effective for that occasional trad project. Whether you call it "sport" or "trad," it's helpful to commit to your goal route (sport or trad) at the beginning of the season, identify the relevant characteristics of the climb (length, angle, difficulty and duration of cruxes, etc.), and adjust your training accordingly to prepare for the specifics of the route.

— Mike Anderson



As routes increase in difficulty, the line between "trad" and "sport" becomes blurred. The standard-setting *East Face of the Monkey*, 5.13d at Smith Rock, OR is the perfect example.  Mike Anderson



Good footwork is a tremendous advantage in trad climbing. Tommy Caldwell using his feet on *Vital Transformation*, 5.12c, Squamish, British Columbia, Canada.  Andrew Burr

size hangboarding, ARCing, Route Intervals, and skill practice (especially outdoors, on relevant rock types). Linked Bouldering Circuits, Campus training, and overhanging indoor climbing should be deemphasized.

Trad climbing is very technical, both in terms of movement and protection skills, and thus more responsive to improvements in technical skill. Therefore, when training for trad goal routes, it is appropriate to emphasize technical skill practice by setting aside extra time for practice. This may necessitate replacing typical indoor training activities with outdoor training, or reducing the volume of indoor sessions to allow for supplemental outdoor sessions, which permit more specific technical skill practice. Base-Fitness training is the easiest to accomplish outdoors. Indoor ARCing should be replaced with Outdoor Mileage (OM) on relevant rock types whenever possible. To maximize the effectiveness of the practice, set skill-development goals, make a plan, and be mindful of these throughout the climbing day. Don't let it devolve into unstructured cragging. Vary the fear and/or risk elements as necessary to stress-proof new skills as they develop.

Regardless of the climbing style — sport or trad — the fundamentals of good climbing technique always apply: weighting the feet, precise and efficient

movement, grip control, breathing, remaining calm, etc. In some cases — when difficult, insecure protection is involved — some of these factors may be *more* important than in sport climbing. Every aspiring trad climber should re-emphasize the learning and practice of these critical abilities. To tailor indoor practice sessions to trad climbing goals, select vertical, or near vertical terrain, and limit footholds to small screw-on holds (aka jibs) and/or sculpted features. It is a common mistake to learn and practice these techniques on indoor face climbs, where the footholds are unrealistically large and positive; then, struggle to apply them later where it counts, on often-slabby outdoor trad climbs with micro-footholds. Therefore, *specifically emphasize transferring these skills to outdoor trad climbs*. This can be accomplished in two ways; first, by utilizing realistic terrain for indoor practice (near-vertical with tiny holds), and second, by setting aside specific outdoor climbing days (such as the OM climbing sessions suggested below) for deliberate practice and stress proofing, as described in **Chapter 3: Skill Development**.

A seasonal training plan is shown on p262 that has been tailored for trad climbing goals. This schedule prescribes more outdoor climbing than the sport climbing-oriented training plans because indoor

climbing gyms are generally not very specific to trad climbing (the typical gym terrain of steep/juggy/slopy doesn't adequately simulate typical trad climbing terrain, rock, and hold types). The following adjustments have been made to tailor this plan to a trad-climbing goal:

- During the Base-Fitness Phase, many of the standard indoor ARC training sessions have been replaced with OM climbing, scheduled to coincide with the weekend. Climbers may adjust as necessary to capitalize on their typical outdoor cragging days.
- The Strength Phase has been lengthened to accommodate outdoor training activities in addition to the nine hangboard workouts. Roped bouldering on hard trad routes or a Limit Bouldering session may be used for these outdoor strength exercises. Also during the Strength Phase, additional Base-Fitness training sessions have been added in the form of OM or ARC training on the days following intense strength training. These adjustments are prescribed in order to enhance the *potential* for trad climbing-specific skill development. That development will only occur if the climber diligently and thoughtfully applies

the principles described in **Chapter 3: Skill Development** toward his or her specific trad-climbing skill weaknesses.

- The Power Phase has been reduced accordingly.
- The Power-Endurance Phase can be absolutely essential or completely superfluous depending on the training goal. This example includes a substantially reduced PE Phase, which exclusively consists of longer-duration Route Intervals to develop the type of fitness required for pumpy trad climbs. Instead of short-duration PE, trad climbers will generally rely on long-duration PE and an increased Maximum Steady State (MSS) resulting from the numerous additional ARC workouts in the season. This modification is suitable for an athlete training for a long enduro crack (climbed mostly below the MSS), or a big-wall free climb with long pitches sporting short, bouldery cruxes.
- The "optional" tag has been removed from many of the Aerobic Exercise workouts to prepare climbers for the workload of multipitch climbing. However, there is at least one full rest day per week.



Where there's a will... Alex Honnold on *No Way Jose*, 5.13c, North Wash, UT.

 Andrew Burr

TRADITIONAL AND BIG WALL FREE CLIMBING

	Week	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Base Fitness	-	<i>Last Week of Performance Phase of Previous Season</i>						
	-	<i>Rest Phase of Previous Season</i>						
	-							*Outdoor Mileage (OM), 8 pitches *Skill acquisition
Strength Phase	1	*OM, 8 pitches *Skill acquisition	*Aerobic Exercise (AE)	*ARC 1: 2 x 20 min *Skill acquisition	*OAE	*ARC 2: 2 x 25 min *Skill acquisition	*Optional Aerobic Exercise (OAE)	*OM, 12 mod pitches *Skill acquisition
	2	*OM, 12 pitches *Skill acquisition	*AE	*ARC 3: 3 x 30 min *Skill practice	*ARC 4: 3 x 25 min *Skill practice	*ARC 4: 2 x 30 min *Skill practice	*OAE	*Outdoor Roped Bouldering (ORB) 1 (hard trad routes)
	3	*OM, 8 mod pitches *Stress-proofing *Skin care	*AE	*Hangboard (HB) 1 *Supplemental Ex	*Opt ARC, 2x20 min *AE *Skin care	*OAE	*HB 2 *SE	*OM, 8 mod pitches *Stress-proofing *Skin care
Power Phase	4	*OAE	*HB3 *SE	*Opt ARC, 2x20 min *AE *Skin care	*AE	*HB 4 *SE	*Opt ARC, 2x20 min *AE *Skin care	*OAE
	5	*ORB 2 (hard trad routes)	*Opt ARC, 2x20 min *AE *Skin care	*AE	*HB5 *SE	*Opt ARC, 2x20 min *AE *Skin care	*OAE	*ORB 3 (hard trad routes)
	6	*OM, 8 mod pitches *Stress-proofing *Skin care	*AE	*HB6 *SE	*Opt ARC, 2x20 min *AE *Skin care	*OAE	*HB7 *SE	*OM, 8 mod pitches *Stress-proofing *Skin care
PE Phase	7	*AE	*HB 8 *SE	*Opt ARC, 2x20 min *AE *Skin care	*AE	*HB 9 *SE	*Opt ARC, 2x20 min *AE *Skin care	*OAE
	8	*LB Outdoors (boulder or routes)	*AE *Skin care	*LB 105 min *SE	*AE *Skin care	*LB 105 min *SE	*AE *Skin care	*LB Outdoors (boulder or routes)
	9	*OM, 8 mod pitches *Stress-proofing *Skin care	*AE *Skin care	*LB 105 min *SE	*AE *Skin care	*LB 105 min *SE	*OAE *Skin care	*RP Attempts *OM, 2 mod pitches
Rest Phase	10	*RP Attempts *OM, 2 mod pitches	*OAE *Skin Care	*WBL 60 min *Linked Bouldering Circ (LBC), DC=2.4	*OAE *Skin Care	*WBL 50 min *LBC, Duty Cycle (DC) = 2.4, SE	*OAE *Skin Care	*RP Attempts *OM, 2 mod pitches
	11	*RP Attempts *OM, 2 mod pitches	*OAE *Skin Care	*OAE	*WBL 40 min *LBC, Duty Cycle (DC) = 2.3, SE	*OAE *Skin Care		*RP Attempts *OM, 2 mod pitches
	12	*RP Attempts (pumpy routes) *OM, 2 mod pitches	*Skin Care	*RP Attempts *OM, 2 mod pitches	*Skin Care	*RP Attempts *OM, 2 mod pitches	*Skin Care	*RP Attempts (pumpy routes) *OM, 2 mod pitches
Performance Phase	13	*Skin Care	*RP Attempts *OM, 2 mod pitches	*Skin Care	*RP Attempts *OM, 2 mod pitches	*Skin Care	*RP/On Sight (OS) Attempts *OM, 2 mod pitches	*Skin Care
	14	*RP/OS Attempts *OM, 2 mod pitches	*Skin Care		*WBL 45 min	*Skin Care		*RP/OS Attempts *OM, 2 mod pitches
	15	*RP/OS Attempts *OM, 2 mod pitches	*Skin Care		*WBL 45 min	*Skin Care		*RP/OS Attempts *OM, 2 mod pitches
Rest Phase	16	*RP/OS Attempts *OM, 2 mod pitches		*OAE, 30 min *Cross-Training		*OAE, 30 min *Cross-Training		*OAE, 30 min *Cross-Training
	17							First Day of Next Season

Key:
 HB - Hangboard Workout LB - Limit Bouldering
 Circuits OAE - Optional Aerobic Exercise
 OM - Outdoor Mileage
 WBL - Warm-Up Boulder Ladder
 DC - Duty Cycle (work-to-rest ratio)
 M/mod - Moderate
 OS - Onsite
 ARC - Aerobic Restoration and Capillarity Training
 m/min - Minutes
 OCT - Optional Cross-Training
 SE - Supplemental Exercises
 RP - Redpoint
 LBC - Linked Bouldering

Seasonal Training Plan for trad route climbing. This example allocates time for outdoor skill development throughout the entire season while emphasizing the Base-Fitness, Strength, and Performance phases and deemphasizing the Power and Power-Endurance phases. Additional Base-Fitness workouts are prescribed through the Strength Phase to further reinforce trad climbing-specific skill development. This schedule offers more opportunities for outdoor climbing than sport climbing-oriented training plans because indoor climbing gyms generally don't simulate trad climbing very well. The activities for each day are listed in chronological order, but the day's primary emphasis is underlined. Note: This schedule includes a two-week-long climbing vacation during weeks 14 and 15 — the ideal time to attempt a trad goal route.

BIG-WALL FREE CLIMBING

Big-wall free climbs and long multipitch climbs entail a whole host of challenges beyond the typical half-pitch crag route. Common characteristics such as scale, remoteness, rock quality, protection, weather, and inaccessibility of crux pitches all conspire to heap difficulties on aspiring climbers. Accordingly, the inherent reward is equally massive, and if success is achieved, it is always worth the effort.

Free big walls are climbed in a wide range of styles, even more so than single-pitch sport climbs. Walls are climbed in every manner from ground-up, onsight, in a day with each team member free-climbing each pitch, to well-rehearsed, weeks-long redpoint sieges (and even topropes) with each pitch free-climbed by only one team member. The choice of style is personal, and won't be debated here. It is only mentioned to point out that the chosen style has an enormous impact on the preparation necessary to execute the climb and time spent on the route. As always, it is best to clearly establish the goal far in

"I have forgotten the monumental amount of work [of the "Free Salathé" campaign]: the painful fingers and hands; the heavy, heavy burden of what, especially at the time, was an awfully tall stack of difficult pitches.... The climb was the greatest breakthrough of our climbing lives. The Salathé experience proved to Todd and me that these walls could actually be free-climbed." — Paul Piana⁵

advance, along with the chosen style, and discuss it honestly between partners in order to properly prepare to achieve the goal. Consider these two similar goals: "to redpoint every pitch of *Moonlight Buttress*" or "to onsight *Moonlight Buttress* in a day." There is a big difference in the training and practice required to prepare for each goal, not to mention the very different logistical options afforded by them.

In addition to the unique characteristics associated with trad climbing in general, the scale and inaccessibility of long multipitch wall routes create several more difficulties that aspiring climbers must overcome. These include logistical and physical challenges, as well as more abstract and unpredictable factors that are part of the game.

LOGISTICAL CONSIDERATIONS FOR BIG-WALL FREE CLIMBING

There are a number of logistical challenges facing aspiring big-wall free climbers that arise from the sheer size and complexity of enormous cliffs. These considerations generally require mental preparation in addition to physical preparation, and range from intricate planning, to fostering the right habits and attitudes, to practicing wall-climbing skills (such as belay setup and gear hauling). These logistical considerations are less dependent on the climber's physical and technical climbing ability, but inattention to them can very easily sabotage an ascent.



Todd Skinner and Paul Piana during the first free ascent of the Salathé Wall, VI 5.13b, El Capitan, Yosemite, CA.  Bill Hatcher

Multipitch Free-Climbing Styles

"I believe people should always strive to do climbs in the best style they think possible, but never criticize others for what they choose to do.... As long as you are not harming the rock or the route for future ascents, and you are honest about what you have done and your style, you should be able to climb in whatever fashion you want." — Tommy Caldwell²

The chosen style of ascent of a big wall is a very important variable that has far-reaching implications on the likelihood of success and the requisite preparation and training. Therefore, it is important to select your style of ascent at the time the goal is established to facilitate further planning. Some relevant characteristics of style that should be considered are given below. At the end of the day, big-wall free climbing is always a team effort, regardless of the chosen style, so agree on the approach and standard of success as a team to avoid wavering commitment or other complications. The various styles generally form a spectrum of possible options that permit a great deal of latitude for creating an achievable plan. *(The format of the list below is not intended to imply a hierarchy of virtue.)*

- What is the standard for "free" climbing (redpoint, pinkpoint, yo-yo, top rope)?
- Must every team member free every pitch?
- Must every team member lead each pitch, or is it acceptable to follow free?
- Are fixed ropes permitted to commute to and from the ground or bivy sites?
- Must the route be climbed in a single push, or is it acceptable to bivy on the ground?
- Must the pitches be climbed in order?
- Is it permissible to pre-rehearse the route?
- Is it acceptable to approach from the top down?
- In the event of a fall, must the entire wall be repeated, or just that pitch?
- Can gear (water, snacks, bivy gear, key pieces of protection) be strategically pre-placed before the climb, or must the climbers carry it on the final ascent?
- Is support from non-climbing team members permitted (belays, gear hauling, etc.)?

There are a wide range of acceptable styles for free climbing a big wall, so don't fall victim to peer pressure or the judgment of armchair critics. The only opinions that matter are those of the climbing team. I've climbed walls in almost every style possible, and I'm reluctant to proclaim one as superior to another — each has its own rewards. Maybe this makes me "unprincipled," but I would instead say "un-fanatical." Personally, I enjoy the variety, and believe each climb warrants a tailored approach. The accessibility of the wall and the amount of time you are able to commit to it are important factors in the decision-making process.

Whichever style you choose, agree on it well before the ascent. Deferring these choices until you are on the route, under pressure in the moment of struggle, is a sure path to later regret. During the goal-setting process, ask yourselves: "What style of ascent will we be satisfied with when the climb is over?" That is the only relevant standard.

— Mike Anderson

LOGISTICAL PLANNING

In the big-wall free-climbing game, logistics can make or break the ascent, so they deserve significant attention. Typically, there will be a limited amount of time for the entire wall-climbing campaign (limited by weather, if nothing else). A lot of work must be accomplished in this time, so upfront planning is critical. The flow of events varies depending on the chosen style, but in general, common steps include:

1. Scout, assess, and plan the route
2. Move gear to the climb
3. Move onto the wall, assess the climbing, and rehearse the crux pitches
4. Prep the wall for an ascent (clean any loose debris; stash food, water, and other supplies)
5. Make the redpoint attempt
6. Repeat steps three through five, as necessary
7. Remove any gear from the wall



Scouting the route is essential to designing a good plan. A high-powered spotting scope like this can reveal useful details that you might otherwise miss.  Andrew Burr

The logistical planning challenge is to complete all of this work in the allotted time, while preserving enough energy, fitness, and motivation for the eventual redpoint. Consider these factors when creating a plan:

- Equipment (haulbags, portaledges, static ropes, etc.)
- Specialty protection (offwidth gear, thin-crack gear, etc.)
- Shoes and clothing (are multiple pairs of shoes or outfits required?)
- Sun/shade exposure and hours of daylight or moonlight
- Optimal time of day to climb crux pitches
- Bivy and rest locations
- Optimal schedule for climbing, hauling, and resting
- Optimal gear-caching sites

Juggernaut

Ascending fixed ropes, or *jugging*, can be an essential skill in the sport of big-wall free climbing. While none of us got into the sport to climb ropes, it's another element that can torpedo an ascent if the climbers don't have a minimum level of competency. Though it is physically exhausting, jugging is a skill that can be learned and improved. The ultimate goal should be to jug efficiently, so as to save precious energy for free climbing. If done properly, the legs should do the majority of the work, especially on less-than-vertical terrain. Much of this comes down to correctly setting up your jugging equipment with tethers and foot stirrups of the proper length. Consult a big-wall instructional book for more details, then get out and experiment with your own gear. Practice jugging over roofs, in corners, along traverses, wearing a pack, or saddled with a heavy rope — all of which may require different tethering arrangements. Jugging on a big wall is rarely straightforward, so be prepared for anything.

One planning approach is to start by estimating how much time will be needed for the final ascent, then budget time for the initial preparation of the climb (i.e., ferrying gear). Any remaining time is allotted for route rehearsal and rest. Finally, prioritize the route's difficulties and plan to spend the most time rehearsing the hardest sections. Factor in the time it will take to reach specific pitches on the wall in order to rehearse them. Utilizing fixed lines, rappelling in from the top, or traversing from an adjacent route should all be considered as means to access the route for rehearsal (in accordance with your chosen style). Keep in mind that route prep, load ferrying, and gear caching can all occur simultaneously with rehearsal, but budget sufficient time for rest amongst these potentially exhausting activities.

"A difficult 5.13 climb thirty feet off the ground is technically no harder than one 3,000 feet off the ground, but tell that to your pounding heart and sweaty palms." — Todd Skinner

PERSONAL GROWTH OPPORTUNITIES

There are certain aspects of big-wall free climbing that can't be controlled. These factors are rarely welcomed by the climbers in the heat of battle, but in the framework of long-term improvement, they can be optimistically thought of as *personal growth opportunities*. They are part of the game — part of what makes it difficult and part of what makes it rewarding. Through careful planning and preparation, it is possible to avoid or mitigate the consequences of these factors:

- **Weather:** Almost by definition, big cliffs come with heightened environmental challenges, including the threat posed by extreme weather. Preparing for the weather is critical because the length of a wall increases the climber's exposure to it, while reducing her ability to react to it. Wall climbers won't have a full arsenal of clothing, and extreme weather conditions (hot or cold) will have more time to wear on the climbers. If the weather turns dangerous, it won't be easy to retreat either. Simple tasks like belaying or setting up a rappel can become onerous in a stiff wind. Try to avoid these issues by planning far in advance to schedule the climb for the best conditions for the chosen cliff (but be flexible once you arrive on the scene). Seek advice from local climbers on the best seasons, and spend some time scouting out the climb and area, if it is unfamiliar to you. Always have an escape plan, and be willing to abandon gear to get down in a hurry. Develop a pitch-by-pitch plan that accounts for sun exposure and time of day to ensure that the hardest pitches can be climbed in the best available conditions.
- **Exposure:** Exposure — the vertical drop below your feet — is another environmental condition brought on by the size of the wall. It can be crippling for some folks, and motivating for others. Frequent, repeated experience is the best way to overcome a negative reaction to exposure and feel more at ease on the high stone. If a specific pitch's exposure interferes with the climber's ability to relax and climb effectively, it can be helpful to reinforce trust in the protection system with progressive *practice falls*. Start by ensuring that the belay is solid, along with the highest piece of protection — and/or the pieces below it too, if necessary. The leader should then hang on the high piece to take the stretch out of the rope and verify the placement, then lower off of it a few feet and take a



Safely navigating poor rock requires attention, patience, and communication. Jason Haas on the first free ascent of *West Side Story*, 5.12c, Fisher Towers, UT.  Andrew Burr

short “toprope” fall. Continue with increasingly longer falls, progressing to short lead falls until you're comfortable with the protection and the exposure, and able to climb freely.

- **Rock Quality:** Big cliffs tend to have worse rock overall than most single-pitch crags. This can contribute to poor protection, the risk of longer falls, and deadly rockfall. Try not to disturb obviously loose rock. This includes avoiding placing protection near loose rock, and making every effort to route the rope away from it. Alert the belayer to any potential for rockfall, and, if possible, wait for him to move under cover before proceeding. A haulbag or pack can be used as a shield if rockfall is inevitable. When faced with a pitch of poor rock, there is no substitute for experience, and some climbers are better than others at managing such chaos. The timid might argue that this is a natural-born talent (or mania) that some climbers have and most do not, but this explanation is simplistic and incomplete. Composure in the face of poor rock is an *intangible* mental skill that can be developed. The following section presents strategies for developing this and other such intangible attitudes that are useful for big-wall free climbing.

“Ultimately, the hardest part of big-wall free climbing is not the exposure, but the acceptance that there is no instant gratification... to gain what we set out to gain, we had to accept our gratification in small parcels and take joy in the process.” — Todd Skinner⁶



Big wall free climbing demands a lot from climbers, including the ability to deal with poor rock, execute unusual moves at a moment's notice, and work closely as a team. Mike Anderson and Rob Pizem on the first free ascent of Angel Hair, V 5.13a, Zion, UT. Keith Ladzinski

ATTITUDES FOR SUCCESSFUL BIG-WALL FREE CLIMBING

Success in big-wall free climbing depends on several intangible abilities. These can be loosely described as the collective mental faculties for persevering through the numerous hardships inherent to wall climbing. These hardships include exposure, loose/dirty/wet/icy rock, vegetation, poor or insufficient protection, mounting fatigue, insufficient sustenance, poor weather, route uncertainty, and non-ideal sending conditions (among many others). The abilities needed to handle these hardships are intangible in that they can't be measured, but it is possible to develop them. Successful big-wall climbers are *robust* — able to perform well under different (often poor) conditions. Like any skill, it is possible to increase a climber's robustness through practice. However, robustness is really a collection of very fundamental *attitudes*, so it's not as simple as learning a new movement skill, like a drop-knee. Attitudes are a result of our past experience and

repeated practices, or habits, which must be cultivated over a lifetime. Therefore, to develop the most robust attitudes, climbers can adopt these habits and reinforce them throughout their career:

- **Urgency:** On a long climb, there is never enough time. The climber can always expect to be under pressure to climb, *and send*, quickly, whether rehearsing pitches in preparation for an eventual redpoint or on the final ascent itself. The team must be highly efficient with all of the logistical tasks such as placing gear, establishing belays, hauling, rappelling, route-finding, and even hiking to the cliff (among many other chores). The more moderate pitches will generally need to be onsighted, in order to save time, so climbers must be competent at onsighting on the given rock type. Crux pitches will need to be assessed, rehearsed, and sent much faster than a pitch at ground level, so climbers can't count on getting complete rest between burns. To improve urgency, climbers should individually practice efficient climbing whenever possible (at the gym and crag, and even in preparation for those activities), and partners should work to develop efficiency as a team. It may be beneficial to practice speed climbing together to develop and refine the logistical systems. To speed the rehearsal and redpointing process, borrow techniques from the sport-climbing game — instead of a stickclip, be prepared to aid a crux pitch or pull on gear to expedite the route-rehearsal process. See "Strategy for a Short Term Redpoint" in [Chapter 13: Redpoint and Onsight Climbing](#) for more tips on quick redpointing.
- **Team Work:** Unlike single-pitch cragging, big-wall climbing is a team sport, regardless of the style of ascent. A good team is essential, and can be built through practice, training, and mutual suffering. Start with a likeminded partner of similar ability, and honestly discuss the team's mutual goals as well as each climber's commitment to them and his ability to spend time to accomplish them — few things are as frustrating as a supposedly "committed" partner with no free time to train or practice. Use the processes outlined in [Chapter 2: Goal Setting and Planning](#) to honestly assess each team member's strengths and weaknesses, assess the weaknesses of the team as a whole, and then make a plan to improve the team. Often the best partnerships include teammates with complementary, rather than over-

lapping, strengths. In many cases, it can be helpful to explicitly define a division of labor or otherwise clearly establish roles and responsibilities, especially on large teams. Very experienced teams will often fall into these roles automatically through habit, but it's always smart to air these subjects in advance. Remember that compatibility and devotion to the team (and its goals) beat physical strength every time.

- **Be the Leader:** While mentors are needed to learn the proper safety systems for climbing early in a career, they can easily become a crutch over time. A more experienced climber who is always picking the routes, leading the harder pitches, hanging the draws, or otherwise “bailing out” the pupil is only stifling the pupil's growth. Once a climber is competent at lead climbing, it's time to step out of the mentor's shadow. An aspiring wall climber should establish her own goals and plans to accomplish them. She should be driving the conversation about which routes to climb, and volunteering to take the sharp end at every opportunity. If she gets in over her head and can't finish a pitch, she should deal with the consequences of dogging, aiding, or leaving gear behind. She should not call in for reinforcements from the former mentor, who won't be available to help on the wall. Instead, she should seek to become self-reliant, push herself on difficult climbs, and practice risk management.
- **Learn to Learn:** Another critical aspect of becoming self-reliant is to personally develop the skill of solving boulder problems. On unfamiliar routes, take the time to experiment with the holds and moves, to solve the problems and learn the beta. Don't take beta from others who won't be there to lend it when it really matters, high on a wall.
- **Get Uncomfortable:** The masses flock to crags that have great rock, inspiring routes, well-textured holds, perfect conditions, and lots of convenience... and why shouldn't they? Climbing is



Chris Lindner enjoying unusually calm weather on *El Nino*, 5.13c A0, Yosemite, CA.

© Brendan Nicholson

supposed to be fun, after all. However, these amenities are uncommon or at least unreliable on a big wall, so special effort should be made to seek out crags and climbs that will expand the climber's tolerance for imperfect conditions. The popular crags that are “nice” to climb at are not the best venues to develop robustness. Instead, seek out the opposite: obscure crags and routes that are generally avoided because of these imperfections. These unpopular, “old school” crags may have stiff grades, runouts, and cobwebs, and they may require unfashionable techniques such as wide jamming and slab climbing. These are the skills needed for big-wall mastery, and the lack of crowds will also require the climber

to be self-reliant in terms of route-finding, beta, and general climbing logistics.

- **Climb Exhaustively:** Every crag has certain routes that are “classic” and receive nonstop attention. These crags also have anti-classics that never get climbed. Strive to climb every route, wherever you are climbing. A big-wall free climb will not afford the option of cherry-picking only the cleanest, most enjoyable pitches. The process of ticking every route at a crag will force the climber to develop a broad repertoire of techniques and confront any weaknesses. It will also help develop route-reading and route-finding skills, both critical to success on a big wall. Furthermore, it will help cultivate an attitude of exploration and a willingness to tread boldly where few others would.
- **Love to Suffer:** The plethora of poor environmental conditions that can arise on a wall are hard to deal with if you’re inexperienced. Rotten or imperfect rock (including wet, icy, dirty, loose, or vegetated stone) is the ultimate example, but poor weather, dehydration, lack of sleep, and fatigue can be equally debilitating. Suffering through similar conditions is the best way to prepare to do it on a wall. Alpine rock climbing* is the best way to get a concentrated dose of all these environmental imperfections, as well as a great way to learn to accept conditions as they are and then deal with them. Wet rock *can* be climbed, and scaling a snowy cliff is a good way to prove it to yourself. In the alpine realm, bad rock is unavoidable, so this is a great place to hone the mental and physical skill of moving past it (just be sure to use a top rope and a helmet for the first forays into the delicate art of “chossaneering”). The beauty of alpine climbing as a training tool is that it is very complementary to sport rock climbing. The typical season for alpinism (late summer) rarely interferes with the prime sport-climbing seasons in North America, and the physical demands don’t overlap with rock-climbing training. Therefore, pile on as much alpine climbing as possible — there is very little reason not to. (*Side effects may include death and dismemberment.)
- **Keep Your Chin Up:** Good morale is essential to any committing adventure. A positive attitude and high spirits can overcome any number of challenges. Cultivate this positive spirit by finding joy in simple things, like the incredible view or

the camaraderie of the team. Endeavor to be habitually upbeat when at the crag or on training climbs. Don’t be a pessimist. Remember that regardless of the situation, you *chose* to be there, and you made that choice because you truly love that type of climbing. Keep your love of climbing at the forefront and embrace challenges as they arise. Big-wall free climbing is often transformative, and such experiences are rarely easy (or completely carefree). Embrace these factors and expect to encounter a few unforeseen obstacles on each adventure. Refer to Ilgner’s *The Rock Warrior’s Way* for more strategies on cultivating a positive mental attitude³.

While transforming one’s climbing attitudes may at first seem overwhelming and impossible, it is possible to make minor day-to-day changes that will add up in the long run. Simply accepting imperfection in daily cragging or training sessions is a good place to start. Don’t be the person at the crag who needs absolute silence, perfect weather, and aligned planets in order to even attempt a redpoint. If the margin of error is that slim on a big wall, failure is certain because all of those factors cannot be controlled and will inevitably be less than perfect on any given day. Instead, resist the urge to control every situation; embrace the chaos, and use it to develop a mental buffer that can eventually be applied on the big wall of your dreams.

PHYSICAL CONSIDERATIONS FOR BIG-WALL FREE CLIMBING

To free-climb an entire wall requires the ability to meet the physical demands of any given pitch, anywhere on the wall. This is a great challenge that is distinct from single-pitch rock climbing. The challenges presented by a big wall may necessitate additional physical training, skill practice, or shrewd planning to improve the body’s coping ability on the route. Some of these considerations are listed below, along with tips climbers can use to better prepare for them. These physical factors should be considered in the context of the specific goal route to determine their relative importance, as well as their likelihood of impacting the climb. If these factors are identified early, a plan can be developed to minimize their impact. In many cases, variations to the training plan may be appropriate, in order to, for example, improve total-body stamina or add/improve a particular movement skill.

- **Stamina:** The volume of climbing required by a big wall is nearly impossible to simulate in train-

ing, and would likely be injurious or counter-productive in the long term. Therefore, during a big-wall ascent, the overarching strategy must be to conserve energy at all times. Logistics will contribute to this, so careful planning is a key coping strategy. On longer climbs, avoid carrying any unnecessary gear, and consider pre-placed caches to conserve energy during the send. Optional cardio work can be added to the training program to prepare for the extra workload of big-wall climbing, without risking overuse injuries to the climbing musculature. Climbing training must take priority, so perform these workouts on climbing rest days, or after the climbing workout on training days.

- **Warming Up:** The crux pitches on long routes must be taken as they come, not necessarily when the climber is ready for them. Often a hard pitch or move will need to be dispatched with little or no warm-up, such as when it is the first pitch off the ground, or the first pitch of the day on a multi-day route. To mitigate this, the climbers can

warm-up on the pitch itself, by pulling through the hardest moves. High on a route, it may be possible to lower down and top rope the previous pitch as a warm-up, or jug a fixed line to get the big muscles firing, if nothing else.

- **Movement Variety:** The sheer amount of rock that the team must overcome guarantees that a wide variety of skills will be needed, while the time urgency imposed by the climb will prevent learning these skills on the spot. Therefore, the climbers need to arrive at the climb already equipped with the preponderance of techniques needed. There is no shortcut to developing a large technique library; it takes a lifetime of climbing. The general strategy of racking up mileage on a wide variety of rock types and styles is the best bet — never pass up an opportunity to learn a new move or address a weakness. For a specific big-wall route, study the route or talk to other climbers to determine if any unique skills are needed — e.g., the ability to jam a certain size crack. If such a skill happens to be a weakness, make time in the training program to improve it.
- **Fatigue:** The demands of multipitch climbing tend to fatigue climbers much differently than the relatively sedentary crag scene. Count on being hungry and dehydrated, then add the whole-body *logistical fatigue* that results from all the ancillary tasks such as:
 - Hiking to the wall
 - Hauling gear
 - Belaying a second from above
 - Hanging belays
 - Rope drag
 - Rappelling
 - And, in the case of a first ascent;
 - Cleaning
 - Drilling bolts

Although food and water are always limited on a long climb, try to stay hydrated and fueled throughout the day, and consider rationing a little extra for the moments leading up to the crux pitches — a favorite snack can be emotionally uplifting in addition to raising blood sugar. Limit sun exposure as much as possible throughout the climb, and don't get too cold either — shivering wastes energy. Budget time for a brief rest immediately before a redpoint attempt on a crux pitch. During this time, spare the leader from any strength-sapping logistical chores, such as hauling gear (to the extent possible). When the time comes, the ability to compose yourself in the face of this fa-



Big wall chores contribute to a level of fatigue unlike that of crag climbing. Mike Anderson hauling gear on Angel's Landing in Zion, UT.

© Andrew Burr

tigue and muster the necessary effort to succeed on a critical pitch is as much mental as it is physical. Logistical fatigue generally doesn't affect the primary climbing muscles, so just because the body *feels tired* doesn't mean it's incapable of cranking on poor handholds. Therefore, try to ignore this tiredness. Apply the same mental techniques that are used for any redpoint and focus on the climbing, not the shouts of protest from your tired body.

PROGRESSIVE LONG-ROUTE TRAINING

Just like any feat, athletes can develop the ability to handle the physical demands of climbing long multipitch routes. The same training principles that were discussed in **Chapter 4: Foundations of Physical Training** apply to long-route training as well. They are:

- Specificity
- Overload
- Recovery
- Reversibility
- Regularity
- Progression
- Variation
- Individualization
- Transfer

To build up to a particular goal route, it can be very beneficial to set intermediate multipitch goal routes that provide physical training, mental training, and skill practice, while also serving as an evaluation of the team's progress. Carefully select intermediate goal routes of appropriate difficulty that mimic the goal route or address key weaknesses, in order to satisfy the principles of effective training. If possible, strive to climb routes on the same rock type or same cliff as your final goal route, and try to tune the length and difficulty of intermediate goal routes to ensure the appropriate progression. Recall that climbing *volume* — the product of intensity and duration (pitch difficulty and overall route length, respectively, in this case) — is the best way to chart progression. Gradually increase the total volume of each intermediate multipitch route throughout the season to build up to the goal route.

The constraints of everyday life will prevent most climbers from regularly climbing long multipitch routes as training for a big wall; in fact, many climbers are lucky to sneak in one long route per season. However, the physical efforts can be simulated at the crag or in the gym by doing climbing mileage days, in which many single-pitch routes are packed into a single day. Using landmark routes as a template (i.e., doing an

Training for a Big-Wall Free Climb

In spring 2004, while living and climbing at Smith Rock, I had the good fortune of meeting the legendary Lynn Hill — the first person to free *The Nose*, and then the first to free El Cap in a day, not to mention her stack of international competition victories. With plans to attempt *Freerider* later that season, I asked what the secret to her big-wall success was, and the answer surprised me.

Lynn relayed that she spent most of her preparation time in France, climbing limestone sport routes. To develop the superhuman endurance necessary for Grade VI-in-a-day free climbing, she would climb 10 to 15 moderate pitches ("moderate" for her, so 5.11 to 5.12), and then finish each day by attempting a few 5.13 pitches once she was already powered down from all the mileage climbing. It was hard to imagine completing such a regimen myself, but I could see how a climber like Lynn Hill, with flawless technique and phenomenal fitness, could pull it off.

In track and field, this would be called "negative split" training, starting off with relatively low-intensity intervals, then progressively dialing up the intensity as fatigue mounts. I applied Lynn's advice to the technical, near-vertical sport routes and stellar cracks of Smith Rock. After several weeks of this training, I set off to meet Mike below the towering walls of Yosemite Valley. Thanks to the solid technical skills I'd developed at Smith, and my newfound levels of stamina, Mike and I walked away a few days later with a team redpoint of *Freerider*.

— Mark Anderson

"Astroman," "Moonlight Buttress," "Half Dome," or "El Cap" day) is a common technique that vastly increases the fun factor and motivation. For example, during a "Moonlight Buttress Day" the climbers might try to complete 10 pitches, including at least six pitches of 5.12- or harder. Include a variety of pitches (in terms of difficulty, style, and quality), just as the goal route will require. The calendar shown in the Figure on p262 includes numerous Outdoor Mileage and indoor ARC training sessions. These workouts can be replaced with progressive long-route training; just be sure to include adequate rest (increasing as the volume increases throughout the season).

Get a Leg Up

Single-pitch rock climbing doesn't require substantial leg strength or endurance, so many sport climbers avoid activities that might beef up their legs and reduce their strength-to-weight ratio. However, multipitch and big-wall projects typically involve plenty of hiking with heavy loads that can cripple you if your legs and lungs are not prepared. Rob Pizem and I learned this the hard way in 2006 when we attempted a first free ascent on Yosemite's Half Dome — a cliff that *starts* 3,000 feet above the valley floor. In the first three days of the trip alone we logged 12,000 vertical feet (and countless miles) of hiking and another 3,000 feet of jugging. The *down-hiking* was the most devastating, and by the fourth day we were absolutely trashed. We had expected to be fatigued, but we were anticipating fatigue of the upper-body and skin variety. Instead, the climb was hinging far more on our ability to ferry loads to the cliff.

We didn't complete the climb that summer for several reasons, so when we made plans to return the following year, we prepared more thoroughly. In addition to my normal climbing training, I made a point to hike up nearby peaks once or twice a week, carrying additional weight in a backpack as I progressed (I hauled rocks or water, and then discarded them for the hike down — a rare geologic phenomenon known as “reverse erosion”). We also carefully planned the climbing and route-prep schedule so that, once in Yosemite, we could get by with less hiking and load hauling in the first few days (we only hiked 7,000 feet of vertical the first three days of the 2007 trip). This two-pronged attack of training and planning solved this key logistics problem, and three weeks later we made the first free ascent of *Arcturus* (5.13b).

— Mike Anderson



If unprepared, the physical toll of big wall free climbing logistics can sabotage an ascent. 📷 Andrew Burr

SUMMARY

- Traditional climbs differ significantly from most sport climbs, but they *do* often resemble vertical face climbs, making such routes good training for trad climbs and vice versa.
- For trad goal routes, climbers should adjust their training to emphasize outdoor-skills practice climbing, Base-Fitness, Route Intervals, and strength training, while deemphasizing Limit Bouldering Circuits and Campus training.
- Long multipitch or big-wall free climbs add a number of challenges that result from the scale, remoteness, and complexity of the climbs. Learn to manage these logistical considerations, and practice good planning on a regular basis.
- For big-wall routes, clearly establish the team's goals well in advance, to include matters of style and the logistical approach. Doing so will clarify the training process needed to achieve the goal.
- Success on a free big wall requires the right attitudes, which can take many years to cultivate. These are habitual, so start early and constantly reinforce them in day-to-day climbing situations.
- Progressive training can be used to prepare for the physical rigors of wall climbing. Develop a plan that produces a steady progression of physical, technical, and mental training.



Melissa Love on *Snake Watching*, 5.13a, Flatirons, CO.  Adam Sanders

BOULDERING



Jake List on *A Maze of Death* V12, The Buttermilks, CA.

📷 Dan Brayack

CHAPTER 15

"I originally perceived bouldering as the distilled essence of the climbing experience, free of extensive paraphernalia and practiced close to the ground." — John Gill⁶

Bouldering is climbing in its simplest form. By eliminating equipment, belayers, and exposure, climbing is simplified to movement over stone. With this freedom, climbers can commit completely to a handful of moves and pursue their true limits. Bouldering is an essential training tool that should be used by all climbers in pursuit of continuous improvement, but it's also an extremely popular and worthy endeavor in itself. This chapter explains how to modify the Rock Prodigy training program to maximize bouldering performance for its own sake (details on utilizing bouldering as part of a larger training program are provided in **Part II: Physical Training**). In addition, the unique mental aspects of bouldering will be discussed, along with strategies for maximizing performance out on the *blocs*.

Bouldering differs from roped route climbing in several important ways, and many of these differences should be accounted for when developing a bouldering-specific training plan:

- **Length:** Short boulder problems require little to no endurance.
- **Hold Type:** Bouldering typically involves more slopers and compression holds than roped climbing.
- **Intensity:** Bouldering moves are much more intense, increasing the importance of strength and power.
- **Duty Cycle:** The work-rest schedule is greatly compressed. Bouldering attempts are much shorter than route attempts, as is the rest period between attempts. Also, when actually climbing, the duty cycle for hand movements is often much faster because of the more dynamic nature of bouldering, and the ability to easily identify and rehearse sequences from the ground.
- **Movement:** Bouldering is often more dynamic than roped route climbing, and many problems entail strenuous opposition moves that require greater core and shoulder strength.
- **Arousal:** The emphasis on powerful and dynamic movement requires high levels of arousal, and the ability to cultivate such a mental state quickly and frequently.
- **Style:** The ability to *hangdog* on routes simplifies the process of discovering and refining beta. This tactic is not typically used in bouldering — you climb from the ground or partway up the problem every time — increasing the effort (and skin) required to unlock sequences.
- **Exposure:** The height and landing area of a boulder problem will affect the consequence of a fall from the problem. These variables produce a spectrum of danger that can range from extremely safe to life threatening.
- **Accessibility:** Proximity to the ground permits many more attempts on a boulder problem per session. This allows difficult sequences to be sent in less time, but also increases the repetitive stress on joints and skin.
- **Group Size:** Roped climbing requires exactly one partner. When bouldering, more is almost always merrier, and a bigger crew means more crashpads, more spotters, and more beta, resulting in a greater willingness and ability to go for it on committing moves.
- **Conditions:** The temperature-sensitive interactions between skin, shoe rubber, micro-crimps, and friction holds are ever magnified in bouldering. Cold is almost always better, and a sub-freezing day on which route climbing would be impossible may be ideal for bouldering. On these days, it takes extra effort to warm up and remain warm throughout the day.